

BE MECHANICAL ENGINEERING

Program Educational Objectives

PEO 1 Become successful Mechanical Engineers who can be competent, innovative and productive in addressing the needs of the Industry or pursue higher education & research in Mechanical Engineering, Marine based Engineering and in other related disciplines.

PEO 2 Grow professionally with their knowledge and be proficient in skills throughout their career.

PEO 3 Demonstrate high standards of ethical conduct, positive attitude and societal responsibilities.

Program Outcome

PO1 Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems

PO2 Problem Analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences

PO3 Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions for complex problems.

PO5 Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice

PO7 Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change

Program Specific Outcome

PSO1 Develop and implement new thoughts on product design and development with the help of modern computer-assisted tools, while guaranteeing effective manufacturing practices.

PSO2 Apply engineering knowledge, design & analysis tools in various problems to resolve issues with the domains of thermal, structural, and fluid mechanics in mechanical and marine based engineering

PSO3 Apply the latest technology of manufacturing and management practices to work professionally in industries or to become an entrepreneur

M.E THERMAL AND FLUID ENGINEERING

Program Educational Objectives (PEOs)

The program educational Objectives of Master's in Thermal and Fluid Engineering is to facilitate the students to:

PEO 1: To train students with good scientific and engineering knowledge so as to comprehend, analyze, design and create novel products and solutions for the real life problems.

PEO 2: To inculcate students in professional and ethical attitude, effective communication skills, teamwork skills, multidisciplinary approach and social values by developing the ability to relate thermal and fluid engineering issues to broader social context

PEO 3: To impart students with in depth knowledge to excel in research and development and to succeed in engineering profession and the life-long learning needed for a successful professional career.

Program Outcomes (POs)

PO1: An ability to independently carry out research/investigation and development work to solve practical problems.

PO2: An ability to write and present a substantial technical report /document.

PO3: An ability to apply advanced concepts of Thermal and Fluid Engineering to design, develop and analyze mechanical systems to render solutions.

PO4: Ability to use advanced techniques, skills and modern scientific and Engineering tools for professional

practice in Thermal and Fluid systems.

PO5: Ability to communicate effectively at all levels of projects and its management and demonstrate leadership qualities in a multidisciplinary scientific research team.

PO6: An ability to engage in independent, reflective, and lifelong learning for the benefits of society.

Program Specific Outcomes (PSOs)

PSO1 Ability to develop and implement new thoughts on fluid mechanics and development with the help of modern computer assisted tools.

PSO2 Ability to apply engineering knowledge, design and analysis tools in various problems to resolve issues with the domains of thermal Engineering.

PSO3 Work professionally in research and development, industry or become an entrepreneur by applying thermal and fluid oriented practices.